

## **PART 4 TECHNICAL REQUIREMENTS**

### **SECTION 1 GENERAL**

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## **SUBSECTION - 1.1 DESCRIPTION OF THE PROJECT**

### **1.1.1. GENERAL**

The Muara Tawar Gas Fired Power Plant consists of the following blocks:

- Muara Tawar Block 1: Combined Cycle Power Plant Multi-Shaft unit, comprising of three Gas Turbines type 13E2, three Heat Recovery Steam Generator units and one Steam Turbine unit.
- Muara Tawar Block 2: two open cycle E class Gas Turbines, type 13E2, a third installed Gas Turbine has been removed to Bali.
- Muara Tawar Block 3: three open cycle E class Gas Turbines, type V94.2
- Muara Tawar Block 4: three open cycle E class Gas Turbines, type V94.2
- Muara Tawar Extension Project Block 5: Combined Cycle Power Plant Multi-Shaft unit, comprising of one Gas Turbine type 13E2, one Heat Recovery Steam Generator units and one Steam Turbine unit.

The Muara Tawar CCPP Add On Project will convert the existing open cycle Block 2, Block 3 and Block 4 to be Combined Cycle Power Plants Multi-Shaft with the following major equipment:

1. Add On to Block 2 comprising of two (2) Heat Recovery Steam Generator units and one (1) Steam Turbine unit which has a capacity of approximately 150 MW at site condition.
2. Add On to Block 3 comprising of three (3) Heat Recovery Steam Generator units and one (1) Steam Turbine unit which has a capacity of approximately 250MW at site conditions.
3. Add On to Block 4 comprising of three (3) Heat Recovery Steam Generator units and one (1) Steam Turbine unit which has a capacity of approximately 250MW at site conditions.
4. Balance of Plant system to support safe and reliable operation of the Add-on CCPP Block-2, 3 and 4. The cooling system for steam turbine condenser will be once through cooling using sea water from existing intake channel.

## **1.1.2. LOCATION OF THE PROJECT**

Muara Tawar Add-On Project is located at the northern part of the district of Bekasi, West Java, approximately 12 km away from Tanjung Priok, the harbor of Jakarta City, towards the east direction.

## **1.1.3. TRANSPORTATION FACILITIES**

### **1.1.3.1 Road**

Access road to Site is available from two directions, i.e. from Tanjung Priok, Jakarta and from Bekasi.

Road conditions from both directions are not adequate for heavy equipment transport to the Site. If the Bidder intends to use the road for equipment transport, they are responsible for the reinforcing of the road.

### **1.1.3.2 Seaport**

The closest available commercial sea port is at Tanjung Priok. There is an unloading facility at the Plant site which could be access through intake canal by low draft barge. Deep draft vessels are not suitable for delivering equipment because of the shallow coastal seabed. It is the Bidder's responsibility to get familiar with sea transport methodology, provide necessary docking and loading equipment and prepare Plant coastal site for transportation and unloading of heavy plant equipment.

### **1.1.3.3 Airport**

The nearest airport, Sukarno - Hatta, is an international airport.

### **1.1.3.4 Hotel**

Excellent hotel accommodations are available in Jakarta.

## **1.1.4. SITE CONDITIONS AND DESIGN CONDITIONS**

The power plant site is located in a tropical area near the sea and will be subjected to a hot, salty and humid climate.

The site conditions data are obtained from Meteorology Station at Tanjung Priok Harbor Jakarta, from year 1983 to 2001, and are summarized as follows. These data are for reference only. The Bidder shall use the most stringent

value of these site condition parameters for the design and performance criteria.

Not with standing the data and information given in this Specification, the Bidder shall thoroughly investigate and familiarize himself with all conditions of the Site and its surrounding area. The Bidder must enquire and satisfy himself as to the physical conditions prevailing at the site and take into consideration all matters and possible contingencies affecting the execution, completion and maintenance of the Works including transport, supply of labour and the effect of the monsoon periods.

Any implications and their expenses arising as a result of the lack of knowledge, understanding and oversight on the part of the Bidder regarding the conditions of the respective Site, shall be to the responsibility and full cost of the Bidder. No condition payment thereof shall be considered nor compensated by the Owner.

Any equipment furnished under this Bid which will be installed outdoor and effected by the Site conditions shall utilize the specified Site design conditions.

#### **1.1.4.1 Rainfall**

- (1) The recorded maximum annual rainfall is 1739.9 mm, and maximum monthly rainfall is 292.2 mm.
- (2) The average annual rainfall is 1327.4 mm.
- (3) The wettest months are generally January, for which the average monthly rainfall is 144.9 mm.

#### **1.1.4.2 Air Temperature**

- (1) The annual average temperature is 28.4°C.
- (2) The maximum temperature is about 31.9°C
- (3) The minimum temperature is about 25.9°C
- (4) Site design Condition: 40°C unless otherwise specified

#### **1.1.4.3 Relative Humidity**

Relative humidity varies from 60.8% to 91.5% and the annual average is about 74.6%.

#### **1.1.4.4 Atmospheric Pressure**

- (1) Maximum : 1010.6 m Bar.
- (2) Minimum : 1008.6 m Bar.

#### **1.1.4.5 Evaporation**

The annual average for evaporation is 2.8 mm/day.

#### **1.1.4.6 Wind Speed**

The maximum recorded wind speed is 4.9m/sec majority to the NorthEast direction.

However, all structures and equipment exposed to wind shall be designed for a basic wind velocity of 120 km/h as per ANSI A58.1.

#### **1.1.4.7 Seawater**

The following seawater data are preliminary and provided to the Bidder for their reference only.

##### **1.1.4.7.1 Seawater Levels**

- (1) Highest High Water Sea Level (HHWL) : EL + 0.94 m
- (2) High Water Tide Sea Level (HWL) : EL + 0.57 m
- (3) Mean Sea Water Level (MWL) : EL + 0.00 m
- (4) Low Water Tide Sea Level (LWL) : EL - 0.57 m
- (5) Lowest Low Water Tide Sea Level (LLWL) : EL - 0.94 m
- (6) Seawater Maximum Temperature : 32.5°C
- (7) Seawater Minimum Temperature : 30.1°C
- (8) Design Condition : 32.5°C

##### **1.1.4.7.2 Seawater Analysis**

Refer to sea water data as stated in attachment in Part 6.

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#### **1.1.4.8 Plant Grade**

The Plant levels are as follows:

|     |                                     |                   |
|-----|-------------------------------------|-------------------|
| (1) | Mean Seawater Level (MWL)           | = EL $\pm$ 0.00 m |
| (2) | Ground level (GL)                   | = EL + 2.80 m     |
| (3) | Main Building 1st floor             | = GL + 0.30 m     |
| (4) | Auxiliary Building 1st floor        | = GL + 0.30 m     |
| (5) | Top of outdoor equipment foundation | = GL + 0.30 m     |

#### **1.1.4.9 Geology**

Preliminary soil data have been performed in the site area from May 5<sup>th</sup>, 2002 until May 22<sup>nd</sup>, 2003 indicates that the subsurface stratigraphy over the site is relatively uniform. Ground water lies within 2 meters of the original ground surface. Very soft to soft silty clays were encountered to a depth of approximately 15 meters. The standard penetration test (SPT) blow counts within this layer are less than 10, between approximately 15 meters and 28 meters depth, the clay is stiff to very stiff with a typical SPT blow count of 15, at approximately 28 meters depth a cemented layer of silts and or/ dense sands is found. The thickness of this layer is ranges from approximately 1 to 7 meters below 28 meters depth with an SPT blow count number greater than 50, below the cemented layer lies stiff to very stiff clay. There is no rock layer within 50 meters depth.

Site investigation also performed at Muara Tawar by the PT Ganesha Piramida in 28 September 2004 until 8 October 2004 at PLTGU block 5. Based on that report, the soil layer can be grouped in various soil conditions according to strength (soil consistency) and grain size. Upper stratum is divided into two (2) soil layers with several sub layers. The marine deposit layers consist of silt layer and clay layer respectively, which are grouped into river alluvium deposits stratum.

The stratigraphy of soil layer is described in Table 1.1-2.

**Table 1.1-2. Sequence of Soil Stratum**

| Soil                    | Depth ( m )             | Layer | Description                                                                                       | SPT<br>N / 30 cm |
|-------------------------|-------------------------|-------|---------------------------------------------------------------------------------------------------|------------------|
|                         | 0.00                    |       | Ground Surface Elevation                                                                          |                  |
|                         | -1.5 ~ -3.5             | 1.a   | Soft to firm, red, CLAY (back filling material) (CH)                                              | up to 6          |
| Marine Deposits         | -6.0 ~ -8.0             | 1.b   | Loose to med. Dense, dark gray, SAND (SP/SM)                                                      | 7 - 26           |
|                         | -10.3 ~ -13.6           | 1.c   | Soft, greenish to dark gray, Silty CLAY (CH)                                                      | 1- 4             |
|                         | -15.0 ~ -17.6           |       | Soft to firm, greenish gray, CLAY (CH)                                                            | 2 - 6            |
|                         | -19.0 ~ -21.0           |       | Stiff, pale brown to gray, silty CLAY/CLAY (CL) with some coral or shell                          | 27 - 34          |
|                         |                         |       |                                                                                                   |                  |
| River Alluvium Deposits | -30.4 ~ -32.0           | 2.a   | Very hard, greenish/yellowish brown, Clayey SILT/SIL (MH/ML)                                      | 30 -54           |
|                         |                         | 2.b   | Very hard, greenish gray, CLAY (CL)<br>Sometime interbeded with Clayey Silt or Silty SAND (MH/SM) | 52 - 69          |
| Old Alluvium Deposits   | drilled up to 37m depth |       | Drill No executed                                                                                 |                  |

### Soil Layer 1.a

This layer consists of soil fill (embankment) which has generally low consistency and various materials. There is red of Silty Clay, but occasionally found coarse material. This layer thickness from existing ground surface up to 3.5 m. N-SPT value is less than 6. This clay has relative high plasticity and high compressible when load is imposed on this soil layer

### Soil Layer 1.b

This layer indicated as marine deposits of well graded Sand, Soil consistency loose to medium dense, dark gray in color with N-SPT value is generally less than 10, but sometime up to 26. The thickness of this layer is about 3 m until 6 m. The sand indicated as a typical shore deposit that has been transported by the river.

### **Soil Layer 1.c**

(1) Upper sub layer

This layer consist of generally very soft to soft of Silty Clay with greenish to dark gray in color. Very low SPT value 1~ 4. The layer thickness is around 3.8 ~ 6.4 m. Highly plasticity and high compressible.

(2) Middle sub layer

The layer consistency getting slightly more high, soft to firm, greenish gray of clay. High plasticity with N-SPT value is 2~ 6. Rock origin indicated as tuffaceous / volcanic ash. The layer has 4 ~ 5.2 m thickness.

(3) Bottom sub layer

The sub soil layer thickness is about 3 m, indicated marine deposits containing shell debris or coral. The layer consistency range is stiff to hard with N-SPT value is around 27 ~ 34. The layer is Silty Clay, pale brown to yellowish gray in color.

### **Soil Layer 2.a**

Well cemented Clayey Silt or Silt with N-SPT value 30 - 54. The soil has a high consistency degree indicated from about 25 m depth. Soil color are greenish to yellowish brown or occasionally pale brown. Soil layer thickness has a range between from 9 m to 12 m thickness. The top level of this layer appears from approximately 25 m depth should be considered as the competent bearing layer for foundation of power plant.

### **Soil Layer 2.b**

The soil layer condition encountered very hard consistency, where the N-SPT values is more than 50, sometime the N-SPT value is up to 69. The soil mass performed by greenish gray to greenish brown good well cemented Clay, this clay layer position is at 30.4 ~ 32 m depth. This layer inserted by another layer namely very hard Clayey Silt (more less 1 m thick only) and very hard Silty Sand.

This layer shall be considerable for foundation of pile, where the N-SPT value is more than 50 with adequate soil thickness. This zone has high skin friction capability for pile.

The project Site is located very close to the shoreline. The ground water level observed in the inshore drillings was encountered at 1 m deep or less.

There is no rock layer found within 50 m under the ground surface.

This geological information is provided for the Bidder's reference only for the preliminary design.

In order to get competent soil parameters for the foundations design, the

Bidder shall perform detail investigations of the soil at the project Site. The scope of investigation shall clarify the soil conditions so that satisfactory and economical foundations can be designed by the Bidder. The scope of soil investigation shall be approved by the Owner.

#### **1.1.4.10 Seismic**

Based on Seismic zone specified in the “Tata Cara Perencanaan Ketahanan Gempa untuk Bangunan Gedung dan Non Gedung (SNI 1726-2012) by Badan Standardisasi Nasional” site location have variable seismic are :

- Peak Ground Acceleration (PGA) = 0.3 – 0.4 g
- Short Period Spectral Acceleration (Ss) = 0.6 – 0.7 g
- 1-sec Period Spectral Acceleration (S1) = 0.25 – 0.3 g

(Last page of Subsection 1.1)

## **SUBSECTION - 1.2 SUMMARY OF WORK**

### **1.2.1. SCOPE OF WORK**

#### **1.2.1.1 General**

Unless stated specifically to the contrary in Schedule 3.2 “Deviations from and Exceptions to Bid Document” where full supporting explanations shall be given about the reason and content of the deviations, the Bidder will be deemed to have agreed fully with the intended requirements of the Bid document.

The Bidder shall be responsible to provide reliable, safe, operable and complete in all respects, the specified Plant within the confine of the specified design and layout.

The Bidder shall include the entire Works as described in or implied by the Bid document. Any matter missing from the Bid document that can be inferred to be obviously necessary for the efficiency, stability and completion of the Works, shall be deemed to be included in the Bid Price.

Works shown in the drawings but not mentioned or described in the Technical Specifications and/or Works described in the Technical Specifications but not shown on the drawings will nevertheless be deemed to be included in this Bid document and their execution is to be covered by the Bid Price.

Similarly, work described in any discipline system specification but not clearly indicated in others discipline system specification will nevertheless be deemed to be included in this Bid document.

Any conflict or discrepancy between technical specification and technical drawing shall be governed / decided by Owner.

The Bidder shall only offer plant and equipment that have been proven in commercial services. Prototype component and equipment are not acceptable.

### 1.2.1.2 Scope of Work

The Work to be performed under this Bid is EPC/ Turnkey Work of:

- a. Converting the two (2) Open Cycle block 2 to Combined Cycle Power Plant by adding two (2) HRSG and one (1) Steam Turbine which has capacity of approximately 150 MW.
- b. Converting three (3) Open Cycle block 3 and block 4 to Combined Cycle Power Plant by adding three (3) HRSG and one (1) Steam Turbine which has capacity in approximately 250 MW per block at site conditions.

The equipment offered shall secure stable and safe operation and be suitable for good integration into the existing facilities.

The Work consists of design, manufactures, inspection and testing at manufacturer's facility, packing for export, shipment, insurance, customs clearance, delivery to Site, unloading, storing, and handling at Site, complete erection, finish painting, on-site testing, and commissioning, performance testing, operation and maintenance during testing and commissioning, warranty and inspection prior to final acceptance of a complete plant as described herein after.

The Bidder shall, at his own cost, obtain all necessary and required permits and licenses for the execution of the Works, for example Building Construction permit (IMB), permit for heavy equipment transportation and road reinforcement, etc.

The above scope of work are summarized but not limited to the following table:

| Item                                                      | Block 2 | Block 3 | Block 4 |
|-----------------------------------------------------------|---------|---------|---------|
| <b>MECHANICAL WORKS</b>                                   |         |         |         |
| Gas turbine by pass damper                                | 2       | 3       | 3       |
| HRSG with and main stack                                  | 2       | 3       | 3       |
| Steam turbine generator unit with water steam cycle       | 1       | 1       | 1       |
| Circulating water system including 2x50% pumps and piping | 1       | 1       | 1       |
| Chlorination plant                                        | 2       |         |         |
| Desalination plant                                        | -       | 2       |         |
| Demineralization Plant                                    | -       | 3       |         |
| Traveling screen with bar screen and rakes                | 2       | 2       | 2       |
| 100/20 tons overhead traveling crane for steam turbine    | 1       |         |         |
| 25/5 tons overhead traveling crane for CW pump            | 1       |         |         |

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|                                                                        |          |       |       |
|------------------------------------------------------------------------|----------|-------|-------|
| 15 tons gantry crane for CW intake                                     | 1        |       |       |
| Miscellaneous cranes and hoists <b>as per M011 clause 1</b>            | 1        |       |       |
| Firefighting detection and alarm system                                | 1 lot    | 1 lot | 1 lot |
| Firefighting system                                                    | 1        |       |       |
| Auxiliary Boiler                                                       |          | 1     |       |
| Fresh water system including water tanks (1500 m³)                     | -        | 1     | 1     |
| Demin water tanks (2000 m³)                                            | -        | 1     |       |
| Potabel water system                                                   | -        | 1     |       |
| Chemical injection system                                              | 1        | 1     | 1     |
| Sampling system                                                        | 1        | 1     | 1     |
| Compressed air system ( service and instrument)                        | 1        | 1     | 1     |
| Laboratory equipment                                                   | 1        |       |       |
| Workshop equipment                                                     | 1        |       |       |
| Pipe rack                                                              | 1        | 1     |       |
| Field piping support and various piping                                | 1        | 1     | 1     |
| HVAC                                                                   | 1        | 1     | 1     |
| Closed cooling water system                                            | 1        | 1     | 1     |
| Hydrogen Plant                                                         | <b>1</b> |       |       |
| <b>ELECTRICAL WORKS</b>                                                |          |       |       |
| Steam Turbine Generator                                                | 1        | 1     | 1     |
| Generator Circuit Breaker                                              | 1        | 1     | 1     |
| Isolated Phase Bus Duct                                                | 1        | 1     | 1     |
| ST Generator Stepup Transformer                                        | 1        | 1     | 1     |
| Unit Auxiliary Transformer                                             | 1        | 1     | 1     |
|                                                                        |          |       |       |
| Non segregated Phase Bus Duct                                          | 1 lot    | 1 lot | 1 lot |
| 6.3 Switchgear                                                         | 1 lot    | 1 lot | 1 lot |
| 400 V PDC and MCC                                                      | 1 lot    | 1 lot | 1 lot |
| PDC Transformer                                                        | 1 lot    | 1 lot | 1 lot |
| Batteries and Chargers                                                 | 1 lot    | 1 lot | 1 lot |
| UPS                                                                    | 1 lot    | 1 lot | 1 lot |
| Lighting and clock system                                              | 1 lot    |       |       |
| Telecommunication system                                               | 1 lot    |       |       |
| CCTV                                                                   | 1 lot    |       |       |
| Grounding system                                                       | 1 lot    |       |       |
| Cathodic protection system                                             | 1 lot    |       |       |
| 500 KV Substation Extension include Substation Automation System (SAS) | 1        | 1     | 1     |
| 500 kV Gas Insulated Line                                              | 1 lot    | 1 lot | 1 lot |
| DCS                                                                    | 1 lot    | 1 lot | 1 lot |
| Emergency Diesel Generator                                             | 1        |       |       |
| Raceways                                                               | 1 lot    | 1 lot | 1 lot |
| <b>CIVIL WORKS</b>                                                     |          |       |       |
| Preparation Work                                                       | 1 LS     |       |       |
| Power House Building and foundation                                    | 1 lot    |       |       |
| Steam Turbine pedestal and aux. foundations                            | 1 lot    | 1 lot | 1 lot |
| HRSG Structure And Foundation                                          | 2 lot    | 3 lot | 3 lot |
| Cooling water (CW) pump pit and foundation                             | 1 lot    | 1 lot | 1 lot |
| CW discharge tunnel                                                    | 1 lot    | 1 lot | 1 lot |
| CW pump house                                                          | 1 lot    | 1 lot | 1 lot |
| CW pipe valve pit                                                      | 1 lot    | 1 lot | 1 lot |
| Firefighting Pump House                                                | 1 lot    |       |       |
| Chlorination plant control building, shed and equipment foundations.   | 1 lot    |       |       |
| Water treatment plant control building and equipment foundations       |          | 1 lot |       |

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-006

|                                                                                        |       |       |
|----------------------------------------------------------------------------------------|-------|-------|
| Desalination plant control building and equipment foundations                          |       | 1 lot |
| Waste water treatment plant control building and equipment foundations and waste ponds | 1 lot |       |
| Control building                                                                       | 1     |       |
| EDG Building and foundation                                                            | 1 lot |       |
| Cable trenches and manholes                                                            | 1 lot | 1 lot |
| Pipe trenches and manholes                                                             | 1 lot | 1 lot |
| 500 KV GIB trench                                                                      | 1 lot | 1 lot |
| 500 KV substation steel structure and foundation                                       | 1 lot | 1 lot |
| 500 KV switchyard control building extension                                           | 1 lot |       |
| 500 KV switchyard pilling work                                                         | 1 lot |       |
| Foudation and firewall for transformer                                                 | 1 lot | 1 lot |
| Tanks foundations (2 units raw water, 1 units make up water)                           | 3     |       |
| Storage room, workshop, laboratory                                                     | 1 lot |       |
| Hydrogent Plant                                                                        | 1 lot |       |
| Road works and finishing works include lighting poles                                  | 1 lot |       |
| Drainage outline structure                                                             | 1 lot |       |
| Fencing                                                                                | 1 lot |       |
| Guard railing                                                                          | 1 lot |       |

The works shall be completed and functioned in accordance with the requirements of this bid documents including but not be limited to the following scope:

#### 1.2.1.2.1 Mechanical Works

##### 1.2.1.2.1.1 Diverter dampers for existing gas turbine bypass stacks in Block 2-3-4 allowing the gas turbines to be operated in simple cycle

As described in Technical Specification M-305.

##### 1.2.1.2.1.2 Heat Recovery Steam Generator

| Item | Description                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Heat Recovery Steam Generator (HRSG) complete with Auxiliary equipment as described in Part 4 Section 3, M-301<br>Complete Water and Steam System as described in Part 4 Section 3, M-301. |
| 2.   | HRSG vents and drains system as described in Part 4 Section 3, M-301 and G-001.                                                                                                            |
| 3.   | HRSG and auxiliaries maintenance equipment as described in Part 4 Section 3, M-301.                                                                                                        |

4. Nitrogen system  
Complete nitrogen system as described in Part 4 Section 3, M-301.
5. Piping systems for the different pressure levels as described in Part 4 Section 3, M-001.
6. Steam blowdown system as described in Part 4 Section 3, M-301
7. Auxiliary steam systems for the different pressure levels as described in Part 4 Section 3, M-304.
8. Turbine By-pass steam systems for the different pressure levels as described in Part 4 Section 3, M-302.
9. Ducting, dampers and stack, as described in Technical Specification M-305.
10. HRSG Steel Structures

All structures and structural steel to support the Heat Recovery Steam Generating Unit and other accessories shall be furnished by the Bidder including but not limited to main columns, girders, floor beams, horizontal and vertical bracings, gallery steel, pipe support steel, other intermediate and auxiliary steel and anchor bolts (see clause 1.2.1.2.3).

#### 1.2.1.2.1.3 Steam Turbines

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Steam Turbines suitable for the different steam inlet pressures in line with the boiler design.                                                                                                                                                                                                                                                                                                                                |
| 1.1  | Steam Turbines with accessories and appurtenances, as described in Part 4 Section 3, M-201, including <ul style="list-style-type: none"><li>• Gland steam sealing system complete with all instrumentation, and gland steam condenser with vapor exhauster.</li><li>• Turbine hydraulic fluid system complete with all instrumentation.</li><li>• Turbine vents and drains system complete with all instrumentation.</li></ul> |

- 1.2 Lubricating Oil System, including:
  - A complete and operable lubricating oil system including lubricating oil conditioner for steam turbine generator and exciter (including all instrumentation) as described in Part 4 Section 3, M-201.
  - Clean and Dirty Turbine Lubricating oil tanks as described in Part 4 Section 3, M-012 and the turbine lube oil purification system as described in Part 4 Section 3, M-209.
2. Condensing plant including:
  - Condenser as described in Part 4 Section 3, M-202
  - Condenser air extraction system including steam jet ejectors and all instrumentation as described in Part 4 Section 3, M-203.
3. Condensate system complete with equipment and piping as described in Part 4 Section 3, G-001.
4. Feedwater system complete with equipment and piping as described in Part 4 Section 3, G-001, including:
  - Boiler feed pumps for the required feedwater pressure levels including drives as described in Part 4 Section 3, M-303
5. Turbine room vents and drains system as described in Part 4 Section 3, G-001, including tanks as described in Part 4 Section 3, M-012 and Pumps as described in Part 4 Section 3, M-006
6. Complete circulating water system with accessories and appurtenances including:
  - Circulating water pumps and drives (complete with local instrumentation) as described in Part 4 Section 3, M-206.
  - Circulating water piping (complete with local instrumentation) as described in Part 4 Section 3, M-208.
  - Circulating water screens complete with Back wash pumps, including screen cleaning system, instrumentation and controls as described in Part 4 Section 3, M-207.

7. Closed Cooling Water System

Complete closed cooling water system as specified in Part 4 Section 3, M-210.

**1.2.1.2.1.4 Common Equipment**

| <b>Item</b> | <b>Description</b>                                                                                                                                                                                                                                                 |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.          | Desalination plant:<br><br>Complete desalination plant system with evaporative type (Multi Stage Flash) as described in Part 4 Section 3, M-440, including but not limited to, complete instrumentation and control system, control panel and its mounted devices. |
| 2.          | Demineralized Water System:<br><br>Complete demineralized water system as described in Part 4 Section 3, M-401.                                                                                                                                                    |
| 3.          | Make up Water System<br><br>Complete make up water system as described in Part 4 Section 3, M-402.                                                                                                                                                                 |
| 4.          | Raw Water System<br><br>Complete Raw Water System & accessories and Domestic Water Supply System for buildings and areas in the Combined Cycle Power Plant as described in Part 4 Section 3, M-403.                                                                |
| 5.          | Waste Water Treatment System<br><br>Complete waste water treatment system consisting of chemical waste water (Neutralization) system as described in Part 4 Section 3, M-404.                                                                                      |
| 6.          | Chemical Injection System<br><br>Chemical injection system feed equipment as described in Part 4 Section 3, M-405 including tanks described in Part 4 Section 3, M-012 and including metering pumps and drives as described in Part 4 Section 3, M-009.            |
| 7.          | Water and Steam Sampling System                                                                                                                                                                                                                                    |

- Complete water and steam sampling system as described in Part 4 Section 3, M-406.
8. Chlorination System  
Chlorination system plant as described in Part 4 Section 3, M-407.
9. Pressurized Air System  
Complete pressurized air system for supply of Instrument Air and Service Air for plant equipment and auxiliaries as specified in Part 4 Section 3, M-410.
10. Hydrogen Plant
11. Gas Fuel System (Not Applicable)
12. Fire Detection Protection and System  
Complete station fire detection and protection system for all building and areas, specified as described in Part 4 Section 3, M-420.
13. Ventilation and Air Conditioning  
Complete ventilation and air conditioning for all buildings and structures as described in Part 4 Section 3, M-421.
14. Central Laboratory  
Complete central laboratory equipment and furniture as described in Part 4 Section 3, M-423.
15. HSD Fuel Oil System (Not Applicable)
16. Auxiliary Boiler refer to part 4 Section 3 M304
17. Cranes  
Overhead traveling cranes, for Steam Turbine Hall and the CW Pump house as described in Part 4 Section 3, M-432.
18. Store equipment and accessories according to M-436.**Not applicable**
19. Miscellaneous Hoists and Trolleys as described in Part 4 Section 3, M-011.

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All mechanical works specified above shall be furnished complete with piping, valves, fittings, support and hangers, insulation and lagging wherever necessary for a complete and functional unit.

#### **1.2.1.2.1.5 Studies to be performed by the Bidder**

The Bidder shall perform engineering studies, calculations and analysis to ensure good mechanical system design at detailed engineering stage. The studies, calculations and analysis shall be submitted according to an agreed Master Document List.

These shall include all studies/calculations/analyses required by individual mechanical specifications, and other studies required by the Owner at detailed engineering stage which in the opinion of the Owner is necessary to obtain a good mechanical system design.

For the requested studies/ calculations/analysis required approval from the Owner, no works can proceed without prior approval from the Owner. The following are the minimum studies/calculations/analysis which shall be performed by the Bidder:

| <b>Item</b> | <b>Description</b>                                                                                                                                             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.          | HP/LP steam piping stress analysis                                                                                                                             |
| 2.          | HP/LP feed water piping stress analysis.                                                                                                                       |
| 3.          | Piping support and hanger calculations.                                                                                                                        |
| 4.          | Equipment capacity and size calculations such as, pump capacity and heat, valve size, safety valve sizing, deaerator and other tank and heat exchanger sizing. |
| 5.          | Water hammer and surge analysis for circulating water system.                                                                                                  |
| 6.          | Analysis for steam demand of auxiliary system.                                                                                                                 |
| 7.          | Noise dispersion study/analysis for the Add-on blocks.                                                                                                         |

#### **1.2.1.2.2 Electrical, Instrumentation and Control Works**

The following scope of work applies for Add On project combined cycle power plant. Any conflict in the contract document, Owner decision is governed.

#### 1.2.1.2.2.1 Generators and Main Generator Output System

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | <p>Generators with accessories and appurtenances as described in Part 4 Section 3, E-101 and as shown on the key single line diagrams including, but not be limited to the followings</p> <ul style="list-style-type: none"><li>• One (1) Steam Turbine Generator for each Block 2, 3 and 4 complete with all accessories.</li></ul>                                                                                                                                                                                                                                                                                                                                                                 |
| 2.   | <p>Generator Auxiliaries, including lubricating oil system as described in Part 4 Section 3, E-101 and appropriate Mechanical Specifications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3.   | <p>One (1) Main Generator Output Systems for each Block 2, 3 and 4 to deliver the output of generator to its generator transformer as described in Part 4 Section 3, E-102 and including, but not be limited to the followings:</p> <ul style="list-style-type: none"><li>• Isolated phase bus duct, including tap-off bus to voltage transformer and surge arrester cubicles, tap-off bus to unit auxiliary transformer and DC field bus duct (if required) and tap-off bus to the static frequency converter equipment (if required).</li><li>• Neutral Grounding Equipment and Bus for generator neutral connection.</li><li>• Voltage Transformer</li><li>• Surge Protection Equipment</li></ul> |
| 4.   | <p>Complete hydrogen supply including all instrumentation and controls, tanks, pumps and piping common to all units and hydrogen supply system specific to each unit including all instrumentation and controls from existing hydrogen storage.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.   | <p>One (1) unit of SF6 Generator circuit breaker for each Steam turbine Block 2, 3 and 4, consisting of isolators and earthing switches, circuit breakers, operating mechanism, all control units, base frame and enclosure as described in Part 4 Section 3, E-103.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |

6. Generator Metering and Protection System complete with all associated equipment as specified and detailed in Part 4 Section 3, E-250.

#### 1.2.1.2.2.2 Plant Electrical

- | Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | <p>Medium voltage auxiliary power distribution system complete with non-segregated phase bus ducts, medium voltage switchgear and associated control, instrumentation, relaying and protection system, as described and specified in Part 4 Section 3, G-001, E-001, E-201, E-202, E-250 and as shown on key single line diagrams including, but not be limited to, the followings:</p> <ul style="list-style-type: none"> <li>6.3 kV switchgear unit auxiliary bus for each Block 2, Block 3 and Block 4 complete with all accessories as specified and detailed in Part 4 Section 3, E-201.</li> <li>Non-segregated phase bus ducts for main feeds to the 6.3 kV switchgear buses and PDC transformers, complete with all accessories as specified and detailed in Part 4 Section 3, E-202.</li> <li>Metering, relaying and protection systems complete with all associated equipment as specified and detailed in Part 4 Section 3, E-201 and E-250.</li> </ul> |
| 2.   | <p>Low voltage auxiliary power distribution systems complete with distribution transformers, low voltage switchgears, motor control centers, and associated control, instrumentation, relaying and protection, and bus transfer schemes as described and specified in Part 4 Section 3, G-001, E-001, E-203, E-204 and E-205 and as shown on Bid Drawings comprising:</p> <ul style="list-style-type: none"> <li>400 V power distribution centers (PDCs) / motor control centers (MCCs) complete with 6.3 kV / 400 V transformers, low voltage switchgear (400 V PDC) and all accessories.</li> <li>230 V power distribution panels complete with all accessories where applicable.</li> </ul>                                                                                                                                                                                                                                                                     |

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3. Station DC power distribution systems complete with batteries, associated battery chargers, DC switchgear and distribution panels, as described and specified in Part 4 Section 3, G-001, E-001, E-220 and E-221 and as shown on key single line diagrams including, but not be limited to, the following:
  - **One (1)** 220 V batteries and **two (2)** chargers complete with mounting racks and all accessories for each ST Generator ADD1 -IIA -005
  - Two (2) 220 V DC distribution panels complete with DC breakers, instrumentation and all accessories for each ST Generator
  - One (1) 24 V battery and associated charger complete with mounting racks and all accessories for each ST Generator, if applicable
  - One (1) 24 V DC distribution panel complete with DC breakers, instrumentation and all accessories for each ST Generator, if applicable
4. Uninterruptible Power Supply System (UPS) complete as described and specified in Part 4 Section 3, G-001, E-001 and E-222 including 230 V uninterruptible power supplies complete with rectifier, auctioneering diodes, solid state inverters, alternate supply regulating transformers, static transfer switches with associated control, instrumentation, and distribution panel complete with all accessories including, but not be limited to, the followings for each ST Generator Unit:
  - **Two (2)** x 100% set of charger/rectifier (one (1) for operation and the other for standby) ADD1 -IIA -006
  - **Two (2)** x 100% Inverters
  - Static Switch
  - Maintenance bypass with isolation transformer
5. Emergency power supply system as described and specified in Part 4 Section 3, G-001, E-001, E-204 and E-223 complete with **one (1)**, 400 V, 3-phase, automatic start, emergency system, cooling equipment, silencer, air filter, fuel oil day tank ADD1 -IIA -007

and piping to the engine, local control panel and all accessories, control instrumentation and alarms.

6. Complete Grounding and Lightning Protection Systems as described and specified in Part 4 Section 3, E-240 and E-241 including:

- Outdoor buried and substructure buried grounding systems, complete with all ground bars, ground rods, cabling and all interconnections, heavy duty compression fittings, exothermic welds, and miscellaneous materials required for the plant areas including substation area as specified in Part 4 Section 3, E-240.
- Electrical grounding of all equipment included in the Bidder's installation and wiring scope including all connections to the underground grounding grids as specified and detailed in Part 4 Section 3, E-241.
- The Bidder shall also provide grounding grid connections and additional ground rods, if required or advised by the Owner, for the plant areas including substation area.
- Complete lightning protection system for all equipment, structures and buildings required in plant areas including substation area as specified in Part 4 Section 3, E-241.

7. Lighting and clock systems, complete with all fixtures, lamps, master and slave clocks, distribution panels, accessories, wiring and conduits, switches and control as described and specified in Part 4 Section 3, E-230, E-231, E-270 and I-200 including:

- Complete lighting system for the plant areas, buildings and structures including substation area.
- Complete outdoor yard and roadway lighting for the plant areas including substation area.
- Complete lighting system for higher structure building and aircraft obstruction lighting for the Stacks in accordance with Part 4 Section 3, E-270.
- Complete clock system for all plant building in accordance with Part 4 Section 3, I-200

8. Plant Communication System consisting of an internal telephone system, public address system, emergency and maintenance telephone system equipment complete with all accessories as described and specified in Part 4 Section 3, E-271 including:
- An internal telephone system consisting of Private Automatic Branch Exchange (PABX) complete with battery and charger, telephone operator's desk, telephone sets, outlet sockets/jacks wired to the PABX, connection to the outside telephone (PT. TELKOM) and line carrier terminal equipment, complete with all accessories, including interface with existing system.
  - Emergency and maintenance telephone system complete with all accessories.
  - Public address system complete with solid state amplifiers, speaker units, microphone stations complete with intercom channels and all accessories for gas turbine, steam turbine and common areas.
9. Cabling Systems
- All medium voltage (MV) power cabling, including terminations, from the switchgear to MV motors and power distribution centers (PDCs) as described and specified in Part 4 Section 3, E-230.
  - All low voltage (LV) power cabling, including terminations, from the PDCs, MCCs and power distribution panels to LV motors, control panels and other LV loads, whether AC or DC, as described and specified in Part 4 Section 3, E-230.
  - All control, instrumentation and communication cables including interconnecting wiring, terminations as specified and detailed in Part 4 Section 3, E-230.
  - All coaxial or fiber optic cables between the power plant central control room and the substation control building for the interfacing between Power Plant and 500 kV Substation.

10. Cable Tray and Conduit Systems.

All cable trays and conduits for power, control, lighting, instrumentation and communication cables required for efficient, economical and convenient wiring and cable routing in the Bidder's scope, in accordance with the design criteria and requirements listed in Part 4 Section 3, G-001, E-001, E-230, E-231, E-232, E-270 and E-271.

11. Underground electrical ducts and trenches for power, control, instrumentation, communication cabling systems and Compressed Gas Insulated Bus (CGIB) or IPB in the plant, and substation areas as specified and detailed in Part 4 Section 3, E-233 or E-102 and as shown on Bid Drawings.

12. Complete Cathodic Protection System to protect the buried piping, the interior of the circulating water piping, condenser water boxes, miscellaneous heat exchangers and all other corrosive portions of circulating water system, pipework and valves as specified and detailed in Part 4 Section 3, E-260 and E-261, required for the plant areas including substation areas.

13. Complete closed circuit television (CCTV) system for observing and security of the power plant and substation including, but not be limited to, the following as described in Part 4 Section 3, E-272:

- Access road
- Entering gate area and the boarding fence
- Inside and outside of steam turbine and gas turbine building.
- Substation area.
- CWP house area.
- Water treatment and desalination plants area.
- Chlorination area.
- Hydrogen generator plant area
- Other area within the power plant and substation considered important for security and observing purposes.

14. Two (2) duplicate LCD consoles shall be provided for monitoring and controlling the security system.

#### 1.2.1.2.2.3 Power Transformers

| Item | Description                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | One (1) Generator Transformer for each Block 2, Block 3 and Block 4 with all accessories as described in Part 4 Section 3, E-301 and E-302..     |
| 2.   | <b>One (1) ST unit aux transformer for Block 2, one (1) ST unit aux transformer for Block 3 and one (1) ST unit aux transformer for block 4.</b> |

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#### 1.2.1.2.2.4 Switchyard (Extension)

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | 500 kV Conventional Substation and appurtenances, as described in Part 4 Section 3, E-400, E-401 as shown on 500kV switchyard extension single line diagrams including, but not be limited to, the following: <ul style="list-style-type: none"> <li>All Connections from Steam Turbine Generator Transformers to the substation as described in Part 4 Section 3, E-401 and E-481 and as shown on Bid Drawings.</li> <li>Shield wires and/or masts to protect the 500 kV Switchyard equipment and connection against direct lightning strikes.</li> </ul> |
| 2.   | 500 kV CGIB and CGIB terminations between the substation and associated EHV connections of Steam Turbine Generator Transformer as described in Part 4 Section 3, E-481 and as shown on Bid Drawings.                                                                                                                                                                                                                                                                                                                                                       |
| 3.   | A complete relaying system for protection of 500 kV equipment as described in Part 4 Section 3, E-440, and as shown in 500kV switchyard extension single line diagrams including, but not be limited to, the following: <ul style="list-style-type: none"> <li>Protective relays for the 500 kV CGIB connection to Steam Turbine Generator Transformer, .</li> <li>Protective relays for the main buses and the diameter</li> </ul>                                                                                                                        |

- buses.
  - Breaker failure protective relays.
  - Protective relays for the branch protection.
4. Substation Control System as described in Part 4 Section 3, E-460 including, but not be limited to, the following:
- Control panels for the local control of the outdoor equipment.
  - Interfacing with the existing SCADA cabinet and all equipment to allow control of the substation equipment remotely from Java Control Center.
5. Instrumentation and monitoring system as described in Part 4 Section 3, I-001, E-460 and E-461, including, but not be limited to:
- Indicating and metering instruments and annunciator for the substation equipment.
  - Oscillograph recording and event recording systems for the substation.
6. The substation AC auxiliary power system is fed from the power plant AC power supply system to meet the requirements of the substation as described and specified in Part 4 Section 3, E-001, E400 and E-450. The system shall include, but not be limited to, the following:
- Existing 400 V AC panels to be adjusted and added.
  - Existing 220 V DC battery and distribution panel to be adjusted and added. Any required 48 V DC system.
  - AC and DC distribution system as described in Part 4 Section 3, E-400 and E-450.
7. The general services including the related instrumentation and control work for the substation and substation extension.
8. All substation steel structures as described in Part 4 Section 3, S-401 and S-402, and bid drawings
9. Foundations for all electric equipment in the power plant and the switchyard supplied by the Bidder including the transformer, structures and the transformer fire walls. The Technical

requirements shall be as described in Part 4 Section 3, S-001.

10. Switchyard extension site work including roads, permanent drainage, sewage treatment system, fencing, landscaping, paving, protective stone fill, field finish painting and galvanizing, etc.
11. Cable ducts for the power and control cables between the substation and power plant. The cable duct shall be as described in Part 4 Section 3, E-233.

#### 1.2.1.2.2.5 Instrumentation and Control

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Hardware, software, engineering and design in accordance with Part 4 Section 3, I-002, I-003, I-005 and functional requirements in accordance with Part 4 Section 3, I-001, I-100 and I-200 to perform the centralized modulating and discrete control functions and plant monitoring function described in System Specification G-001 including, but not limited to, the following items:                                                                                                                  |
| 2.   | HRSG Control, Protection, and Monitoring System including: <ul style="list-style-type: none"><li>• HP and LP Drum Level Control System</li><li>• Steam Temperature Control System</li></ul>                                                                                                                                                                                                                                                                                                                 |
| 3.   | Steam Turbine Control, Protection, and Monitoring System including: <ul style="list-style-type: none"><li>• Steam Bypass Control System</li><li>• Main Steam Pressure Control System</li><li>• Turbine Oil Control System</li><li>• Vacuum and Gland Seal Control System</li><li>• Condensate Control System</li><li>• Deaerator Control System</li><li>• Steam Turbine Control System</li><li>• Steam Turbine and Generator Control System</li><li>• Steam Turbine and Generator Tripping System</li></ul> |

- Steam Turbine Supervisory System
4. Coordination Distributed Control System:
- Load Frequency Control
  - Plant Auto Start-up, Loading and Shutdown
  - Plant Protection System
5. Ancillary Plant Control and Protection System:
- Desalination Plant Control and Protection System
  - Circulating Water Pump and Screen Control System
  - Closed Cooling Water Control System
  - Condenser Tubes Cleaning and Back Washing Control System
  - Water Treatment Plant Control System
  - Waste Water Treatment Plant Control System
  - Plant Fire Protection Control System
  - Chlorination Plant Control System
  - Pressurized Air Control System
6. Plant Monitoring System and Human Machines Interface
- Monitoring, analysis and computation for machinery vibration, plant performance and efficiency alarms, sequence of event record, process variables.
  - Logging for long term and short term including alarm, daily, weekly, demand, post trip, sequence of event and special.
  - VDU displays for the monitoring and the control system including bar charts, graphics, trends overviews, group and alarm summaries.
  - VDU displays using information transmitted via communication data link for Common Ancillary Plants such as water treatment, desalination, chlorination plants, etc. This information may also be used for logging function.
  - Control Desk and Panels as specified in Part 4 Section 3, I-100 including all instrumentation mounted on those

desk and panels.

#### **1.2.1.2.2.6 Studies to be performed by the Bidder**

The Bidder shall perform engineering studies, calculations and analysis to ensure good electrical and I&C system design at detailed engineering stage. The studies, calculations and analysis shall be submitted according to agreed master document list.

These shall include all necessary studies/calculations/analysis required by individual electrical / I&C specifications, and other studies required by the Owner at detailed engineering stage which in the opinion of the Owner is necessary to obtain a good electrical / I&C system design.

For the requested studies/ calculations/analysis required approval from the Owner, no works can proceed without prior approval from the Owner.

| <b>Item</b> | <b>Description</b>                                                                                                                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.          | Short circuit study of all electrical systems including all LV systems and DC Systems, in his Scope of Works.                                                                                 |
| 2.          | Voltage-drop and Voltage-dip Study of all electrical systems in his Scope of Works.                                                                                                           |
| 3.          | Cable Sizing Calculation for all cables and wires in accordance with IEC - 60287.                                                                                                             |
| 4.          | Sizing calculation for all electrical equipment in his Scope of Works, including Busducts, Transformers, Switchgear, Batteries, Diesel Generator, etc. in accordance with relevant standards. |
| 5.          | Protection and relay setting & coordination study for all electrical equipment in his Scope of Works.                                                                                         |
| 6.          | Insulation Coordination Study for all electrical equipment in his Scope of Works.                                                                                                             |
| 7.          | Grounding Complete Design Calculations                                                                                                                                                        |
| 8.          | Lightning Protection Complete Design Calculations.                                                                                                                                            |
| 9.          | Cathodic Protection Complete Design Calculations.                                                                                                                                             |
| 10.         | Lighting Complete Design Calculations.                                                                                                                                                        |

11. Power System Stability Study to confirm the excitation/governing system parameters.
12. Complete Structural Calculations for
  - Bus Ducts including IPB
  - Cable Trays
  - Towers
  - Gantry
  - Compressed Gas Insulated Bus (CGIB) and other structures for electrical equipment.
13. Specific Studies
  - Electromagnetic interference to Instrumentation & Control and communication circuit in substations and power plant.
  - Effects of short circuits, out of step, faulty synchronization and reclosing on the Turbine Generator shaft lifetime.
14. Sizing of Control valves
15. I/O calculation (study for computer capability based on I/O mapping)

#### **1.2.1.2.2.7 Studies and calculations**

All studies and calculations required by individual electrical and I&C specifications, and other studies according to agreed master document.

#### **1.2.1.2.3 Structure, Civil and Building Works**

##### **1.2.1.2.3.1 General**

The Bidder shall furnish and install the Structure, Civil and Building Works, complete as specified in the Part 4 Section 3, S-001 "General Design Requirement of Structure, Civil and Building Works" and in other technical specifications and shown on the approved drawing.

The Bidder shall, at his own cost, arrange for the Owner all permits and licenses required for construction of the Structure, Civil and Building Works

(e.g. IMB/Ijin Mendirikan Bangunan, utilization of land or sea area for disposal material etc.)

The Structure, Civil and Building Works shall include, but not limited to, the following principal elements

Surveying and Setting Out, Soil Investigation, Field and Laboratory Testing, Site Preparation, Piling, Foundation, Dike Wall, Super Structural, Architectural and Building Facility Works for HRSG, Steam Turbine Building, Central Control Building, Transformer, Circulating Cooling Water System Structures, Cooling Pump House, Chlorination Plant, Cable Duct, All Auxiliary Plants, Warehouse, and all associated Civil Works.

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#### 1.2.1.2.3.2 Scope

The scope of Structure, Civil and Building Works are detailed in Part 4 Section 3, S-001 General Design Requirement of Structure, Civil and Building Works in details. Therefore, in this sub-section, the scope thereof is itemized only.

The Bidder shall design, Construction, furnish all materials, labor, equipment, supplies, supervision, transportation, delivery to the site, unloading, storing and handling at site, erection and testing of the following item:

| Item  | Description                                                                                                                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Site Investigation                                                                                                                                                                                                                 |
| 1.1   | Surveying and Setting Out                                                                                                                                                                                                          |
| 1.2   | Soil Investigation                                                                                                                                                                                                                 |
| 1.3   | Field and Laboratory Test                                                                                                                                                                                                          |
| 2.    | Site Development                                                                                                                                                                                                                   |
| 2.1   | Site Preparation                                                                                                                                                                                                                   |
| 2.1.1 | Provision of temporary facilities                                                                                                                                                                                                  |
|       | <ul style="list-style-type: none"> <li>• Temporary site office for Owner and Engineer</li> <li>• Temporary site office for Bidder</li> <li>• Temporary access road and parking</li> <li>• Temporary water supply system</li> </ul> |

- Temporary electric supply system
  - Temporary drainage ditches
  - Temporary fence between existing Plant area and project area
  - Preparation of Laydown area
  - Survey of existing underground structures and utilities
- 2.1.2 Land reclamation including stripping, grading and leveling for final ground level
- 2.1.3 The structures, foundations, roads, fence, drainage ditches, etc. which are located in the Project area and required to be removed. Disposal of demolished debris and materials shall be responsible for the Bidder at the account of the Bidder
- 2.2 Permanent Road and Parking Lot
- 2.3 Permanent Drainage and Ducts
- Storm Water Drainage
  - Sanitary Water Drainage
  - Oily Water Drainage
  - Waste Water Drainage
- 2.4 Fence and Gate
- Piling (if necessary)
  - Excavation & backfilling
  - Concrete foundation
  - Fence and gate
  - Other related work
- 2.5 Power Plant Site Finishing and Environmental Protection
- Excavation & Backfilling
  - Landscaping
  - Other related work
3. Pile Loading Test
4. Structural, Architectural and Building Facility Work

- (a) Piling and Foundation
  - Driving pile
  - Pre-boring/Pre-augering (if required)
  - Settlement monitoring during construction.
- (b) Foundation Works
  - Excavation, dewatering, foundation preparation, backfill and compaction.
  - Lean concrete and concrete mat/structure.
  - Formwork and reinforcing steel.
  - Miscellaneous embedded steel.
  - Anchor bolts.
  - Grouting
- (c) Structural Works
  - Fabrication of steel structure.
  - Erection of steel structure.
  - Bolting and welding connection.
  - Gallery works.
  - Metal decking works.
  - Concrete structure.
  - Coating and painting works.
- (d) Architectural Works
  - Roof
  - Interior and exterior wall
  - Windows, louvers and doors
  - Finish painting works
- (e) Building Facility Works
  - Water supply system for building
  - Drainage system for building
  - Sanitary fixture

- Power supply system for building (refer to E-001)
- Communication system for building (refer to E-271)
- Lightning protection system with grounding for building (refer to E-241)

#### 4.1 **HRSG Area**

##### 4.1.1 Heat Recovery Steam Generator Foundation (HRSG)

- Piling
- Concrete mat Foundation, excavation and back fill
- Concrete Structure
- Elevator structure
- Architectural Works
- Other related works

##### 4.1.2 Stack for HRSG

- Piling
- Concrete mat Foundation, excavation and back fill
- Other related works

##### 4.1.3 Miscellaneous Equipment Foundation Required in Area of HRSG

- All miscellaneous equipment pad and pedestal foundation above ground level.

#### 4.2 **Steam Turbine Area**

##### 4.2.1 Steam Turbine Building

- Piling
- Concrete foundation, excavation and back fill
- Structural Works
- Metal Siding and Roofing
- Gallery Works
- Architectural Works

- Building Facility Works
  - Modify Existing Building Wall and Roof
  - Drainage Works
  - Other related works
- 4.2.2 Steam turbine (ST) and Generator Pedestal
- Piling
  - Concrete mat foundation, concrete pedestal, excavation and back fill
  - Anchor bolt
  - Grouting
  - Embedded material
  - Other related works
- 4.2.3 Central Control Building for Block 1 was designed to accommodate the control equipment for Block 2 Add On control equipment. A new Control Building for Block 3 and 4 has to be built.
- Piling
  - Concrete Foundation, excavation and back fill
  - Structural Works
  - Concrete / Concrete block wall / Brick wall
  - **built-up roofing**
  - Architectural Works
  - **Gallery Works**
  - Building Facility Works
  - Connection corridor
  - Electrical work
  - Other related works
- 4.2.4 Miscellaneous Equipment Foundation Required in Area of Steam Turbine
- All miscellaneous equipment pad and pedestal foundation above ground level.

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4.3            **Electrical Structures**

4.3.1          Generator Transformer and Auxiliary Transformer. Foundation for Steam Turbine.

- Piling
- Concrete Foundation, excavation and back fill
- Firewall
- Drainage
- Oil Pit
- Gravel pavement
- Fencing
- Other related works

4.3.2          Isolated Phase Bus Duct Structures for Steam Turbine Building to Switchyard

- Piling
- Concrete foundation, excavation and back fill
- Structural Works
- Other related works

4.3.3          Switchyard

- Piling (if required)
- Concrete foundation, excavation and back fill
- Structural Works
- Cable trenches
- Drainage and pit
- Fence with gate
- Maintenance road
- Other related works

4.3.4          Underground Cable Ducts And Cable Trenches

- Piling (if necessary)
- Concrete foundation, excavation and back filling.

- Structural Works
  - Trench cover
  - Sump pit
  - Other related work
- 4.3.5 Miscellaneous Equipment Foundation Required in Area of Electrical structure
- All miscellaneous Equipment pad and pedestal foundation above ground level.
- 4.4 **Cooling Water Circulating System.**
- 4.4.1 Cooling Water Pump House and Pump Pit
- Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Gallery Works
  - Architectural Works
  - Building Facility works
  - Other related work
- 4.4.2 Cooling Water Intake System
- Modification works of the existing Intake structure
  - Cooling water pump pit including traveling screen
  - Extension of existing cooling water pump house to accommodate CW pumps for Blocks 2, 3 and 4
  - Gantry crane foundation
  - Cooling water pipe foundation
  - Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Metal siding and roofing
  - Architectural Work

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
|       | <ul style="list-style-type: none"> <li>• Gallery Works</li> <li>• Building Facility Works</li> <li>• Modify Existing Building Wall and Roof</li> <li>• Other related Works</li> <li>• Earthworks for installation of cooling water supply pipe</li> </ul>                                                                                                                                                                                                                                                                        |                      |
| 4.4.3 | Cooling Water Discharge Tunnel & Channel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                      |
|       | <ul style="list-style-type: none"> <li>• Piling</li> <li>• Concrete foundation, excavation and back fill</li> <li>• Structural Works</li> <li>• <b>Modification existing open channel</b></li> <li>• Other related works</li> </ul>                                                                                                                                                                                                                                                                                              | ADD2<br>-IIA<br>-002 |
| 4.5   | <b>Foundation and Structure for BOP</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
| 4.5.1 | Chlorination Plant:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |
|       | <ul style="list-style-type: none"> <li>• Chlorination Building (refer to Clause 4.8)</li> <li>• Salt &amp; sodium hypo chlorite storage tank</li> <li>• Piling</li> <li>• Concrete foundation, excavation and back fill</li> <li>• Structural Works</li> <li>• Gallery Works</li> <li>• Architectural Works</li> <li>• Building Facility Work</li> <li>• Concrete wall and dike (if required)</li> <li>• Drainage and pit (if required)</li> <li>• Fencing surrounding the storage tank</li> <li>• Other related work</li> </ul> |                      |
| 4.5.2 | Desalination Plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |
|       | <ul style="list-style-type: none"> <li>• Piling</li> <li>• Desalination Control Building</li> <li>• Concrete foundation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |                      |

- Structural Concrete Works
- Water tank foundation
- Acid storage tank foundation
- Excavation and back fill
- Other related works

#### 4.5.3 Water Treatment Plant

- Piling
- Water Treatment Control Building
- Water tank foundation
- Acid storage tank foundation
- Concrete foundation, excavation and back fill
- Structural Works
- Gallery Works
- Architectural Works
- Building Facility Works
- Other related works.
- Concrete wall and dike (if required)
- Drainage and pit (if required)

#### 4.5.4 Waste Water Treatment Plant

- Primary metal cleaning waste water pond
- Secondary metal cleaning waste water pond
- Concrete foundation, excavation and back fill
- Piling
- Concrete foundation, excavation and back fill
- Waste Water Treatment Control Building
- Structural Works
- Architectural Works
- Building Facility works

- Other related works
- 4.5.5 Outdoor Piping Support and Foundation
  - Piling (if necessary)
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Other related works.
- 4.6 Outdoor Equipment Foundation
  - Piling (if necessary)
  - Concrete foundation, excavation and back fill
  - Roof and structure (if necessary).
  - Construction machinery foundation
  - Other related work
- 4.7 **Miscellaneous Buildings and Structures**
- 4.7.1 Emergency Diesel Generator Building
  - Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - **Concrete / concrete block wall / brick wall**
  - Gallery Works
  - Architectural Works
  - **Built-up roofing**
  - Building Facility works
  - Other related work
- 4.7.2 Water Treatment Control Building
  - Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Gallery Works
  - Architectural Works
  - Building Facility works
  - Other related work

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- 4.7.3 Waste Water Treatment Control Building
- Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Gallery Works
  - Architectural Works
  - Building Facility works
  - Other related work
- 4.7.4 Desalination Control Building
- Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Gallery Works
  - Architectural Works
  - Building Facility works
  - Other related work
- 4.7.5 Storage Building for B3 Waste and Material
- Piling
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Gallery Works
  - Architectural Works
  - Building Facility works
  - Other related work
- 4.7.6 Laboratory Building, workshop, storage room
- Piling
  - Concrete foundation, excavation and back fill
  - Structural Works

|       |                                                                                                                                                                                                                                                                                                         |              |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|       | <ul style="list-style-type: none"> <li>• <b>Concrete / concrete block wall / Brick Wall</b></li> <li>• Gallery Works</li> <li>• Architectural Works</li> <li>• Building Facility works</li> <li>• Other related work</li> </ul>                                                                         | ADD1-IIA-010 |
| 4.7.7 | <p>Firefighting pump house</p> <ul style="list-style-type: none"> <li>• Piling</li> <li>• Concrete foundation, excavation and back fill</li> <li>• Structural Works</li> <li>• Gallery Works</li> <li>• Architectural Works</li> <li>• Building Facility works</li> <li>• Other related work</li> </ul> |              |
| 4.7.8 | <p>Hydrogen Building</p> <ul style="list-style-type: none"> <li>• Piling</li> <li>• Concrete foundation, excavation and back fill</li> <li>• Structural Works</li> <li>• Gallery Works</li> <li>• Architectural Works</li> <li>• Building Facility works</li> <li>• Other related work</li> </ul>       |              |
| 4.8   | <p>Workshop.</p> <p><b>To be deleted.</b></p> <p><b>Already included at item 4.7.6</b></p>                                                                                                                                                                                                              | ADD1-IIA-011 |

- 4.9 Guard House
- Piling (if required)
  - Concrete foundation, excavation and back fill
  - Structural Works
  - Architectural Works
  - Building Facility works
  - Other related work

#### **1.2.1.2.3.3 Studies to be performed by the Bidder**

The Bidder shall perform engineering studies, calculations and analysis to ensure good structure, civil and building design at detail engineering stage which shall be submitted to the Owner for review and approval.

These shall include studies/calculations/analysis according to agreed master document list required which is necessary to obtain a good structure, civil and building engineering design. The studies include provision of the existing plant modification to Combined Cycle Power Plant.

For the requested studies/ calculations/analysis required approval from the Owner, no work can proceed prior the approval from the Owner.

Following studies/calculations/analysis shall be, but not limited to, performed by the Bidder during construction.

| <b>Item</b> | <b>Description</b>                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------|
| 1.          | Soil Investigation, settlement analysis from in-situ laboratory test, and other required data for detail engineering process |
| 2.          | Pile capacity analysis with study of negative skin friction effect                                                           |
| 3.          | Concrete mixture analysis                                                                                                    |
| 4.          | Storm drainage system analysis and calculation.                                                                              |
| 5.          | All foundation design analysis and calculation.                                                                              |

6. All building and structural design analysis and calculation including stack, crane support, etc.
7. Architectural conceptual design.
8. Construction method study
9. Steel structure material test including welding test
10. Air circulation for room and building.
11. Air conditioning and ventilation calculation for all room and buildings as required
12. CW flow test simulation or a modeling test
13. Other studies related to the Civil Work

**1.2.1.2.4 Operation and Maintenance Training for Owner's Personnel**

Operation and Maintenance Training for Owner's personnel at manufacturer's work and at site is described in Part 4, Subsection 2.13.

**1.2.1.2.5 Supply of Spare Parts**

Supply of Spare Parts as described in Part 4, Subsection 2.11

**1.2.1.2.6 Maintenance Tools and Test Equipment**

Maintenance tools and Test equipment as described in Part 4, Subsection 2.11.

**1.2.1.2.7 Supply of Consumables**

Supply of all consumables required during erection, commissioning test up to COD of the respective Block including initial charge of all lubricating oils and greases, fuses, indicating lamps, filters, lubricants, chemicals and condenser tube cleaning balls, etc.

The items and quantities shall be listed and submitted with the offer.

**1.2.1.2.8 Manufacture and Shop Test Inspection**

Manufacture and Shop test inspection for Owner's personnel.

**1.2.1.2.9 Plant Inspection at the end of Warranty Period**

Plant inspection at the end of warranty period prior to final acceptance.

**1.2.1.2.10 Design, Operating, Maintenance and Systems-Manuals**

Design, Operating, Maintenance and systems manuals for all systems and equipment supplied, as described in Part 4, Subsection 2.10

**1.2.1.2.11 Reports, Documents and As Built Drawings**

Reports, Documents, as-built drawings on completion for record purposes, etc.

**1.2.1.2.12 Arrangement of Permit Letter Application**

Arrangement of Permit Letter application from the Government of Indonesia Concerning:

- Building Construction Permit (IMB)
- Heavy Equipment Transport
- Other later Permit Letter related to the Project

**1.2.1.2.13 Detailed Calculations, Studies and Analyses**

Submission, according to agreed master document list, of detailed calculations, studies and analyses necessary for sizing, designing, performance, etc. for the following works:

- Mechanical
- Electrical and Control & Instrumentation
- Civil and Structural

**1.2.2. WORKS AND SERVICES NOT INCLUDED**

The following items are not part of this Bid:

- For Block 2-3-4: Supply of necessary fuels and electrical power for Testing and Commissioning refer to part 4 section 4, subsection 4.8
- Operators to operate the equipment during Testing and Commissioning

**1.2.3. TERMINAL POINTS AND INTERFACES**

- The Bidder shall be deemed to have inspected and examined the existing Power Plant and to have satisfied himself as to the terminal points and interfaces of existing equipments, pipings, structures and other facilities.

All necessary interfacing works for the proper function of the Combined Cycle Power Plant system shall be deemed to be in the Bidder's scope.

- The Bidder shall be deemed to have obtained all necessary technical data and information (including civil, mechanical & electrical) for Bidder's own design purpose, as to risk, contingency and all other circumstances, which may influence or affect to tender.
- It shall be the Bidder's full responsibility to ensure that the ends of piping, plant and equipment in the Bidder's Scope of Work match up to the connecting piping, plant or equipment in both physical dimensions and chemical composition.
- All termination works except those otherwise defined in the Bid Document shall be the responsibility of the Bidder.

#### **1.2.3.1 Mechanical Works**

The terminal and interfacing points of the Bidder's Scope of Works are:

- Flue gas duct from existing gas turbine flue gas of block 2 (GT 2.1 & 2.2), block 3 (GT 3.1, 3.2 & 3.3) and block 4 (GT 4.1, 4.2 & 4.3).
- HRSG pit sump piping of block 2 to existing waste water treatment plant of block 1 & 2
- Fire protection system to existing fire system of block 2, 3 & 4
- Seawater
  - Intake: existing intake channel
  - Outfall: existing common outfall channel for Block 1 to 5.
- Waste water: Existing common out fall channel for block 1 to 5

#### **1.2.3.2 Electrical and I&C Works**

The terminal and interfacing points of the Bidder's Scope of Works are as follows:

#### **1.2.3.2.1 Interfacing Gas Turbine Control**

The Bidder shall perform the connection work to the existing gas turbine control. All work and accessories for this interface and necessary to make proper functioning of the system are Bidder responsibility.

#### **1.2.3.2.2 500 kV Switchyard**

The Bidder shall perform the connection work to the existing 500kV switchyard and shall integrate the 500 kV switchyard of this Bid into the existing 500kV switchyard.

Additional equipment such as relay panel, mimic panel, battery interface panel, fault recorder, motors, etc. required for the operation of the 500kV switchyard shall be installed in the existing substation control building. All work and accessories for this interface and necessary to make proper functioning of the system in diameter 6, 11 and 12 shall be included in the scope.

#### **1.2.3.2.3 SCADA**

The Bidder shall wire all signals required for SCADA to the SCADA interface panel and other equipment to be provided by the Bidder and to be installed at the substation building. All required transducer, cabling, routing and other accessories shall be provided.

#### **1.2.3.2.4 AC/DC Supply for Switchyard Extension**

The Bidder shall provide connect his AC/DC supply to the existing AC/DC switchgear at the Substation Control Building.

#### **1.2.3.2.5 Grounding**

The Bidder shall connect the grounding grid to the existing grounding grid in accordance with the technical specification.

#### **1.2.3.2.6 Raceway**

In general, the Bidder shall provide his own raceway for his cable. Bidder's cable entering the existing facilities may use the existing raceway with written approval from the Owner. When the existing raceway is not sufficient, the Bidder shall provide additional raceways himself.

#### **1.2.3.2.7 Other**

The Bidder responsibility to connect and integrate to the existing system of other necessary system such as CCTV system, Telecommunication system, control system etc, such that complete and proper system is fulfilled.

#### **1.2.3.3 Structure, Civil and Building Works**

The terminal and interfacing points of the Bidder's Scope of Works are, but not limited to, followings.

##### **1.2.3.3.1 Road**

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If additional roads are required (either temporarily or permanently), Bidder shall include so.

##### **1.2.3.3.2 Pavement with Grading**

- Existing gravel pavement with grading at switchyard

##### **1.2.3.3.3 Storm Water Discharge Ditch**

- Existing storm water drainage ditch located at the above road and storm water is drained to the existing outfall. New ditch shall be connected to the existing.

##### **1.2.3.3.4 Intake Water Open Channel**

- Existing intake water open channel at north side

##### **1.2.3.3.5 Discharge Outfall**

- Existing discharge outfall channel

##### **1.2.3.3.6 Fence at Switchyard**

- Existing fence at switchyard at north and south side

(Last page of Sec 1.2)